

EQUATIONAL PROOFS OF JACOBSON'S THEOREM

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ABSTRACT. A classical theorem by Jacobson says that a ring in which every element x satisfies the equation $x^n = x$ for some $n > 1$ is commutative. According to Birkhoff's Completeness Theorem, if n is fixed, there must be an equational proof of this theorem. But equational proofs have only appeared for some values of n so far. This paper is about finding such a proof in general. We are able to make a reduction to the case that n is a prime power p^k and the ring has characteristic p . We then prove the special cases $k = 1$ and $k = 2$. The general case is reduced to a series of constructive Wedderburn Theorems, which we can prove in many special cases. Several examples of equational proofs are discussed in detail.

1. Introduction

Jacobson has shown in [Ja45] that every ring in which every element satisfies an equation of the form

$$x^n = x$$

is commutative, where $n > 1$ may depend on x . This theorem has been strengthened by various authors. For example, Herstein [He53] showed that it is enough to require that $x^n - x$ is central. Recently, Anderson and Danchev [AD20] proved the commutativity of rings satisfying the equation $x^n = x^m$, where n, m are of opposite parity.

In the present paper, we are interested in the special case of rings satisfying $x^n = x$ for a fixed number $n > 1$, which we call n -rings. The case $n = 2$ corresponds to the well-studied boolean rings, for which there is a simple equational commutativity proof (see Example 2.4). More generally, by Birkhoff's Completeness Theorem [Bi35] (see also [Ta79] for a modern account), there must be an equational proof of the commutativity of n -rings, which thus exists of simple symbolic manipulations and substitutions that derive $xy = yx$ from the ring axioms and $x^n = x$.

However, Jacobson's proof is not equational, and indeed quite sophisticated. More simple proofs of Jacobson's Theorem have been published by Forsythe and McCoy [FM46], Herstein [He54, He61], Rogers [Ro71] and Dolan [Do76], but arguably the most elementary and slick proofs have been given by Wamsley [Wa71] and Nagahara and Tominaga [NT74] (their proofs are the same). Still, these proofs are not equational.

Equational proofs have only been found for some values of n so far. Moreover, they can be quite challenging to find. Buckley and MacHale [BM13] present the classical case $n = 2$, several elementary proofs in the case $n = 3$, and also some useful lemmas and generalizations. Wavrik [Wa99] discusses the cases $n = 3, 4$, several similar commutativity problems, and also presents a general algorithm which is supposed to derive $xy = yx$ from $x^n = x$ for general n , but the author does not verify that it always works. Zhang [Zh90] presents an algorithm for even n , which, in numerical experiments, often reduces the problem to the cases $n = 2$ and $n = 4$. (The present paper shows that this approach is too optimistic, though.)

Morita's impressive work [Mo78] gives equational proofs for even numbers ≤ 50 (except for 16, 22, 32, 46, which are much harder) and odd numbers ≤ 25 . Some larger numbers are covered as well. MacHale [Ma86] covers the case that n is the sum of two distinct proper powers of 2, even under the weaker assumption that $x^n - x$ is central. Buckley and MacHale

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have continued this study in [BM12]. Forsythe and McCoy [FM46] give an equational proof that p -rings of characteristic p are commutative, where p is a prime number. The general case, however, seems to be an open problem, and no systematic approach has been taken so far.

If we follow Birkhoff's proof in our case, we get the following evidence for the existence of an equational proof: Consider the free ring in two variables $\mathbb{Z}\langle X, Y \rangle$, consisting of non-commuting polynomials in X and Y . Then the quotient ring $\mathbb{Z}\langle X, Y \rangle / \langle f^n - f : f \in \mathbb{Z}\langle X, Y \rangle \rangle$ is an n -ring. By Jacobson's Theorem, this ring is commutative. In particular, the image of the commutator $XY - YX$ vanishes. Hence, there must be an equation

$$XY - YX = \sum_{i=1}^s g_i \cdot (f_i^n - f_i) \cdot h_i \quad (1)$$

for $f_i, g_i, h_i \in \mathbb{Z}\langle X, Y \rangle$ for $i = 1, \dots, s$. Once it has been found, this non-commutative polynomial equation can be verified from the ring axioms alone. Evaluating this equation in any n -ring yields an equational proof that it is commutative, since the right hand side vanishes.

However, that approach does not tell us at all how f_i, g_i, h_i look like explicitly, and the existing proofs of Jacobson's Theorem do not *construct* f_i, g_i, h_i . In fact, the proofs use non-constructive methods such as the axiom of choice (with the exception of [Wa71, NT74, Do76]) and the law of the excluded middle (LEM).

The LEM states that for every statement P , either P or $\neg P$ holds. Equivalently, $\neg\neg P$ implies P . For example, notice that [Wa71] merely shows $\neg\neg P$ when P is the statement that n -rings are commutative. This also means that Jacobson's Theorem has not been proven so far in intuitionistic logic, which is a type of constructive logic that does not include LEM [Da01].

For an equational proof, only constructive methods are allowed, and we need to *find* an equation like (1). On the other hand, such a single equation is quite complicated in practice, and an equational proof should better be divided into multiple steps that are easier to motivate and to digest. For example, the equation

$$\begin{aligned} XY - YX &= ((X + Y)^2 - (X + Y)) - (X^2 - X) - (Y^2 - Y) \\ &\quad + ((YX)^2 - (YX)) - ((-YX)^2 - (-YX)), \end{aligned}$$

in $\mathbb{Z}\langle X, Y \rangle$ proves that every 2-ring is commutative. But it is better to present the proof in a more structured way (as in Example 2.4). The case of 3-rings illustrates this even better (see Remark 2.13).

Our goal is to develop a method that works for every value of n . We also prove several reduction results, which generalize observations from [Mo78, Ma86, Zh90]. They are motivated by the well-known classification of commutative n -rings, which we recall in Section 3.

Our first result is a reduction to the case of prime characteristic. For reasons to be explained in Remark 3.10, we write $p = 0$ instead, which also includes the zero ring.

Theorem A (Theorem 3.14). *Let $n > 1$ and assume that for every prime number p with $p - 1 \mid n - 1$ there is an equational proof that every n -ring with $p = 0$ is commutative. Then there is an equational proof that every n -ring is commutative.*

The reader might object that this theorem is true since Jacobson's Theorem in conjunction with Birkhoff's Theorem shows that there is an equational proof anyway. But we have already observed that this approach does not *construct* a proof. The actual meaning of our theorem is that its proof is a method that, given equational proofs with the additional assumption $p = 0$, outputs an equational proof of the general case. The goal of the theorem is not to verify that a certain statement is true in the classical sense. Its goal is to construct equational proofs. And of course, our theorem does not assume Jacobson's Theorem to be true.

The next result concerns a useful reduction to prime powers.

Theorem B (Theorem 4.1). *Let $n > 1$ and let p be a prime number with $p - 1 \mid n - 1$. Let k be the lcm of all $d \geq 1$ such that $p^d - 1 \mid n - 1$. Then there is an equational proof that every n -ring with $p = 0$ is a p^k -ring.*

Hence, in order to give an equational proof that n -rings are commutative, one may restrict to p^k -rings with $p = 0$. Again, the actual content of the theorem is the method that results from its proof, which we then apply in several examples. For example, every 9-ring with $5 = 0$ is a 5-ring, and every 10-ring is a 4-ring. It is not easy to prove this directly without any guidance, but our proof is the outcome of a general method, so that no individual ideas are necessary to find it.

Morita [Mo78] proved several instances of Theorem 4.1: Direct and clever calculations show that n -rings coincide with 2-rings for $n = 6, 12, 14, 18, 20, 24, 26, 30, 38, 42, 44, 48$, that n -rings coincide with 4-rings for $n = 10, 28, 34, 40$, that n -rings coincide 8-rings for $n = 36, 50$, and that 46-rings coincide with 16-rings. (Some of these calculations are omitted, though.) In contrast, our approach does not depend on a specific value of n . Incidentally, the classification of the numbers n such that n -rings with $2 = 0$ coincide with 2-rings can be found in [HLY94].

The next result concerns the special case $k = 1$.

Theorem C (Theorem 5.5). *Let p be a prime number. If A is p -ring with $p = 0$, then every element of A can be explicitly written as a $\mathbb{Z}$ -linear combination of idempotent elements. In particular, A is commutative by an equational proof.*

The last conclusion comes from the basic fact that in a reduced ring, idempotent elements are central. We also provide a version of that theorem that works over any finite field.

Sometimes the case $k = 1$ is enough: Let us call a natural number $n > 1$ *simple* if every prime power q with $q - 1 \mid n - 1$ is actually a prime number. There are many simple numbers, the first ones are 2, 3, 5, 6, 11, 12, 14, 18, 20, and for these, Jacobson's Theorem is easy to prove (Theorem 5.2). We can turn it into an equational proof as well:

Theorem D (Corollary 5.7). *For every simple number $n > 1$ there is an equational proof that every n -ring is commutative.*

Even though the recipe is always the same, these equational proofs are not “uniform” in n , though. They have to be written down for every single value of n , and for large values of n this can be quite cumbersome. We do not know if there is a uniform proof.

For non-simple numbers, our method is often able to reduce the problem to smaller numbers. For example, the commutativity of 2023-rings can be reduced to the commutativity of 4-rings (see Example 5.12).

We also give a proof of the case $k = 2$, inspired by Morita's proofs for $n = 9, 25$.

Theorem E (Theorem 6.4). *Let p be a prime number. There is an equational proof that every p^2 -ring with $p = 0$ is commutative.*

In conjunction with the previous reduction results, this covers many more examples. For example, we find an equational proof that 73-rings are commutative (Example 6.5).

As for the general case, we take inspiration from the elementary proof of Jacobson's Theorem in [Wa71, NT74]. It reduces the problem to finite rings, where Wedderburn's Theorem comes into play, which states that finite division rings are commutative. We are able to make the reduction constructive and even equational. We just have to replace Wedderburn's Theorem by the following constructive variant: If $f \in \mathbb{F}_p[T]$, then $W_{p,k,f}$ states that in a p^k -ring with $p = 0$, we have

$$ba = f(a)b \implies ba = ab.$$

In classical logic, Wedderburn's Theorem implies $W_{p,k,f}$ (Lemma 7.3). We then prove: